

1. Master Framework Shell Structure

The core workspace of the application runs within a three-pane structural layout using a JavaFX BorderPane system. This composition maximizes layout stability, handles scaling across various monitor resolutions natively, and secures role-based data encapsulation.



2. Workspace Subsystem Breakdown

- **Top App Header (HBox):** Hosts identity fields, active module titles, and the localization system. Contains a global toggle to bind interface assets to Albanian or English resource structures.
- **Navigation Workspace (VBox):** Positioned on the left boundary. This sidebar reads user permission configurations post-authentication and dynamically controls accessibility visibility to clinical or billing screens.
- **Dynamic Content Stage (StackPane):** The center control hub. Swaps viewport controls dynamically via the parent layout wrapper, ensuring lightning-fast load intervals below 3 seconds (**NFR-01**).
- **System Status Bar (HBox):** Locked to the bottom boundary. Provides real-time technical statuses including database sync health, active communication tasks, and system timestamps.

3. Core View Implementations & Security Behavior

3.1 System Access Portal (LoginForm)

A centered layout featuring input validation parameters. If consecutive verification failures breach the safety limit of 5 attempts, user fields lock out instantly, triggering an admin-unlock restriction alert banner to block automated attacks (**NFR-07**).

3.2 Patient Workspace Hub (PatientManagerView)

A split interface layout uniting registration forms and live lookup tables. Evaluating a patient's date of birth triggers an immediate automated classification engine, formatting the account structure into a **Pediatric**, **Adult**, or **Senior** workflow subclass (**FR-26**).

3.3 Clinical Workspace (DoctorConsultationView)

A multi-pane command board presenting critical data metrics in parallel viewsets. Features active allergy and drug interaction scanners that generate blocking error overlays during prescription configurations to prevent critical medical errors (**FR-42**).

3.4 Technical & Financial Dashboards

Integrates the Lab Results Portal (highlighting abnormal metabolic levels), the Invoice Engine (generating secure billing entries), and the Executive Oversight Dashboard tracking system-wide logs and data mutation audit paths (**NFR-08**).

4. Architectural Data Integrity Rules

All clinical inputs (diagnoses, laboratory results, prescriptions, and immunizations) implement immutable property parameters. Once submitted to the system core, editing functions lock down permanently (`setEditable(false)`), enforcing non-repudiation and timestamp compliance required by health laws (**NFR-21**).